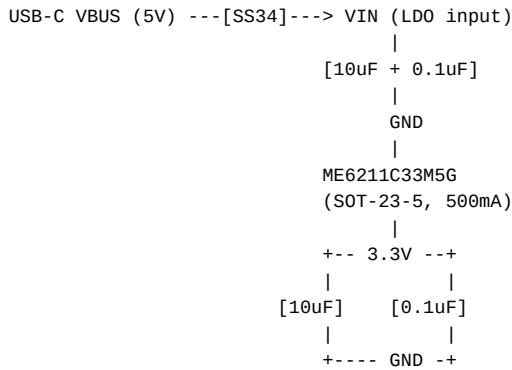


# PC Monitor - Schematic Diagram

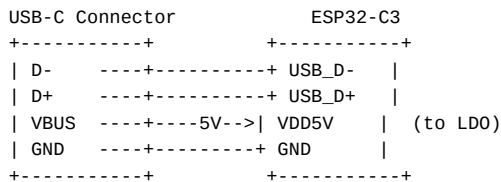
ESP32-C3-WROOM-02-N4 + 1.54" ST7789 LCD + DHT11

## 1. POWER - USB 5V to 3.3V



3.3V rail feeds: ESP32-C3 module + LCD + DHT11

## 2. USB SERIAL/JTAG (Built-in, no external chip)



Note: No CH340/CP2102 needed! ESP32-C3 has built-in USB Serial/JTAG peripheral.

## 3. ESP32-C3-WROOM-02-N4 Pin Connections

| Left Side (Pins 1-18) |        |          |  | Right Side (Pins 19-36) |        |           |  |
|-----------------------|--------|----------|--|-------------------------|--------|-----------|--|
| +-----+               |        |          |  | +-----+                 |        |           |  |
| Pin                   | Signal | Use      |  | Pin                     | Signal | Use       |  |
| +-----+               |        |          |  | +-----+                 |        |           |  |
| 1                     | GND    | PWR      |  | 19                      | GPI019 | ---       |  |
| 2                     | GPI02  | LCD_BL   |  | 20                      | GPI020 | ---       |  |
| 3                     | GPI03  | DHT11    |  | 21                      | GPI021 | LCD_RST   |  |
| 4                     | GPI04  | LCD_DC   |  | 22                      | VDD33  | 3.3V      |  |
| 5                     | GPI05  | LCD_CS   |  | 23                      | EN     | 10K+0.1uF |  |
| 6                     | GPI06  | LCD_MOSI |  | 24                      | VDD33  | 3.3V      |  |
| 7                     | GPI07  | LCD_SCK  |  | 25-29                   | NC     | ---       |  |
| 8                     | GPI08  | 4.7K UP  |  | 30-36                   | GND    | PWR       |  |
| 9                     | GPI09  | 4.7K DN  |  |                         |        |           |  |
| 10                    | GPI010 | ---      |  |                         |        |           |  |
| 11                    | GPI011 | ---      |  |                         |        |           |  |
| 12                    | GPI012 | ---      |  |                         |        |           |  |
| 13                    | GPI013 | ---      |  |                         |        |           |  |
| 14                    | GPI014 | ---      |  |                         |        |           |  |
| 15                    | GPI015 | ---      |  |                         |        |           |  |
| 16                    | GPI016 | ---      |  |                         |        |           |  |
| 17                    | GPI017 | ---      |  |                         |        |           |  |
| 18                    | GPI018 | ---      |  |                         |        |           |  |
| +-----+               |        |          |  | +-----+                 |        |           |  |

GPI00/GPI01: Used internally by flash (NOT available)  
GPI08: 4.7K to 3.3V (strap: HIGH = normal boot)  
GPI09: 4.7K to GND (strap: LOW = normal boot)

## 4. LCD Interface (ST7789 1.54" SPI / ILI9341 2.8")

| LCD Pin | Signal | ESP32-C3 GPIO |
|---------|--------|---------------|
| -----   | -----  | -----         |
| 1       | GND    | GND           |
| 2       | RST    | GPI021        |
| 3       | DC     | GPI04         |
| 4       | MOSI   | GPI06         |

|   |     |                         |
|---|-----|-------------------------|
| 5 | SCK | GPI07                   |
| 6 | VCC | 3.3V                    |
| 7 | CS  | GPI05                   |
| 8 | BL  | GPI02 (or 100R to 3.3V) |
| 9 | GND | GND                     |

Backlight: GPI02 -> 1K -> NPN(2N7002) -> LED-  
 Alternatively: tie BL to 3.3V via 100R  
 (GPI02 briefly high at boot, acceptable)

## 5. DHT11 Temperature/Humidity Sensor

| DHT11 Pin | Signal | ESP32-C3                      |
|-----------|--------|-------------------------------|
| -----     | -----  | -----                         |
| 1 (left)  | VCC    | 3.3V                          |
| 2         | DATA   | GPI03 (+ 4.7K pullup to 3.3V) |
| 3         | NC     | (not connected)               |
| 4 (right) | GND    | GND                           |

Note: 4.7K-10K pullup required on DATA line.  
 Keep wire short (<20cm) for reliable readings.

## 6. Reset / Strapping Circuit

EN (Pin 23):  
 +3.3V --[10K]--+-- EN ---- ESP32-C3  
                   |  
                   [0.1uF]  
                   |  
                   GND

GPI08: +3.3V --[4.7K]-- GPI08  
 GPI09: GPI09 --[4.7K]-- GND

USB-C connector for power + programming:  
 VBUS -> SS34 -> LDO input  
 D- -> ESP32-C3 USB\_D-  
 D+ -> ESP32-C3 USB\_D+  
 GND -> system GND  
 CC1/CC2: 5.1K to GND (if using USB-C)

# Bill of Materials (BOM)

| Item | Qty | Part                 | Package       |            |
|------|-----|----------------------|---------------|------------|
| 1    | 1   | ESP32-C3-WROOM-02-N4 | 36-pin module | Main M     |
| 2    | 1   | ME6211C33M5G         | SOT-23-5      | 3.3V       |
| 3    | 2   | 10uF MLCC            | 0805          | Power      |
| 4    | 2   | 0.1uF MLCC           | 0603          | HF         |
| 5    | 1   | 1.54" LCD (ST7789)   | module        | SPI 24     |
| 6    | 1   | DHT11                | 4-pin T0      | Temp+Hu    |
| 7    | 1   | 4.7K resistor        | 0603          | DHT        |
| 8    | 2   | 4.7K resistor        | 0603          | GPI08 u    |
| 9    | 1   | 10K resistor         | 0603          | E          |
| 10   | 1   | SS34                 | SMA/SOD-123   | Reverse po |
| 11   | 1   | USB-C connector      | 16-pin SMT    | Power      |
| 12   | 1   | 0.1uF MLCC           | 0603          |            |
| 13   | 1   | 2N7002 / A03400      | SOT-23        | BL MOSF    |
| 14   | 1   | 1K resistor          | 0603          | BL ga      |
| 15   | 1   | PCB                  | custom        | Doubl      |

## Notes:

- All resistors 0603 package, +/-5% tolerance
- All capacitors 0603/0805 MLCC, 16V or higher
- PCB: FR4, 1.6mm, double-sided, HASL finish
- Antenna area on PCB: keep clear of ground plane
- USB-C VBUS current: 500mA minimum recommended